Predicting medication-associated altered mental status in hospitalized patients: Development and validation of a risk model

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Purpose. This study presents a medication-associated altered mental status (AMS) risk model for real-time implementation in inpatient electronic health record (EHR) systems.

Methods. We utilized a retrospective cohort of patients admitted to 2 large hospitals between January 2012 and October 2013. The study population included admitted patients aged ≥18 years with exposure to an AMS risk–inducing medication within the first 5 hospitalization days. AMS events were identified by a measurable mental status change documented in the EHR in conjunction with the administration of an atypical antipsychotic or haloperidol. AMS risk factors and AMS risk–inducing medications were identified from the literature, drug information databases, and expert opinion. We used multivariate logistic regression with a full and backward eliminated set of risk factors to predict AMS. The final model was validated with 100 bootstrap samples.

Results. During 194,156 at-risk days for 66,875 admissions, 262 medication-associated AMS events occurred (an event rate of 0.13%). The strongest predictors included a history of AMS (odds ratio [OR], 9.55; 95% confidence interval [CI], 5.64–16.17), alcohol withdrawal (OR, 3.34; 95% CI, 2.18–5.13), history of delirium or psychosis (OR, 3.25; 95% CI, 2.39–4.40), presence in the intensive care unit (OR, 2.53; 95% CI, 1.89–3.39), and hypernatremia (OR, 2.40; 95% CI, 1.61–3.56). With a C statistic of 0.85, among patients scoring in the 90th percentile, our model captured 159 AMS events (60.7%).

Conclusion. The risk model was demonstrated to have good predictive ability, with all risk factors operationalized from discrete EHR fields. The real-time identification of higher-risk patients would allow pharmacists to prioritize surveillance, thus allowing early management of precipitating factors.

Keywords: altered mental status, electronic health records, prediction model, risk score

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Altered mental status (AMS) encompasses a spectrum of abnormal neuropsychiatric disorders of mentation with various time courses and severities. AMS is an independent predictor of increased length of hospital stay, morbidity, mortality, and higher cost of care.[1–5] AMS frequently represents delirium, which is characterized by inattention, cognitive impairment, and alteration of consciousness.[6] Upon hospital admission, the prevalence of delirium is estimated to be 10% to 31%, and the rate of new-onset delirium is estimated to be 3% to 29%.[7] The wide range of prevalence estimates reflects broad variability in screening tools, diagnostic criteria, and patient populations.[8]

The etiologies of AMS are multifactorial, often the pathophysiological consequence of acute medical illnesses. Many risk factors are modifiable and susceptible to intervention, such as electrolyte abnormalities, infections, impairments of vision or hearing, and sleep disturbances. Importantly, numerous medications such as benzodiazepines, opioids, and corticosteroids are known to independently induce or exacerbate AMS.[9–11] Less-than-optimal pharmacotherapy, therefore, represents an important modifiable risk factor amenable to pharmacist intervention.[9,12] The prevention and management of AMS requires a multidisciplinary approach in which pharmacists play an essential role, particularly in evaluating for medication-related causes of AMS. A pharmacist evaluation results in opportunities for intervention including reducing precipitating medications, evaluating for medication withdrawal by assessing medication history, optimizing pain regimens, correcting electrolyte abnormalities, and optimizing antibiotic regimens. Tools are needed to help prioritize and target resources to increase the likelihood of successful intervention.

The identification of patients at highest risk for AMS can enable prioritization of resources and ultimately improve clinical outcomes. Protocols directed at reducing risk factors for AMS have demonstrated improvements in outcomes, but we did not identify any driven by pharmacists.[12–14] We aimed to construct and validate a dynamic risk model for hospitalized patients who receive AMS risk–increasing medications (ARIMs) for real-time use in an inpatient electronic health record (EHR). This study is part of a larger project to build an automated EHR-based scoring system that ranks hospitalized patients at high risk for preventable adverse drug events.[15–17]

KEY POINTS

- A model for evaluating medication-associated altered mental status risk derived exclusively from electronic health record data was demonstrated to have excellent performance characteristics.
- The strongest predictors included in the model were a history of altered mental status, alcohol withdrawal, history of delirium or psychosis, presence in the intensive care unit, and hypernatremia.
- Dynamic predictive models can facilitate focused interventions in populations at higher risk for medication-induced altered mental status events.

Methods

Study population. We utilized a retrospective cohort of hospitalized patients from 2 University of Florida (UF)–affiliated hospitals: UF Health Shands (852 beds) and UF Health Jacksonville (695 beds). We extracted inpatient EHR data from patients admitted between January 1, 2012, and October 31, 2013 (UF Health Shands), and between March 1, 2013, and October 31, 2013 (UF Health Jacksonville), using Epic Systems Software (Epic, Verona, WI).

Patients included in the study were aged ≥18 years. We confined the study population to patients who had exposure to an ARIM within the first 5 hospitalization days to predict AMS events that were likely attributable to medication exposure. Drug monographs found in Clinical Pharmacology[18] and Micromedex[19] were used to identify ARIMs. Monographs containing key words for any type of AMS event (i.e., agitation, aggressiveness, confusion, coma, delirium, hallucination, irritability, lethargy, psychosis, and restlessness) as an adverse event (rather than indication) were further categorized into risk categories based on reported rates (Appendix A). ARIMs include opioids, benzodiazepines, barbiturates, antihistamines, anticholinergics, antidepressants, and other AMS risk medications. ARIM exposure information was extracted from inpatient electronic medication administration records.

Risk model development was planned for each of the first 5 hospital days with ARIM exposure; a hospital day of ARIM exposure was the unit of analysis. Hospital days beyond 5 days were omitted because of small sample sizes. We considered development of separate models for each day to accommodate changes in the contribution of risk factors to predicted risk as patients progressed during their hospitalization or new risk-factor information became available. Models predicted the occurrence of AMS on the following day using information from the ARIM-exposed day (modeling day) and any previous days as appropriate. Thus, for a prediction day to be considered, patients had to be in the hospital at least until the end of the modeling day.

Patients with baseline dementia, with a positive toxicology screen, or on mechanical ventilation were excluded given the presence of a strong non–medication-associated causal pathway for developing AMS. We also excluded patients receiving a continuous infusion of sedative or paralytic agents within the first 5 hospitalization days. Patients remained in the study cohort after having had AMS if they again met inclusion criteria (e.g., AMS event occurred on day 2 and AMS reoccurred on day 5 of the same admission).

AMS outcome. AMS events were identified by a measurable mental status change (as assessed using a validated assessment scale or alternative measure and defined per study criteria; Table 1) documented in the EHR in conjunction with administration of an atypical antipsychotic or haloperidol. We developed the operational AMS definition through review of definitions of AMS that had been used in published literature and discussions with a technical expert panel composed of experts in medication safety and clinical pharmacy.[15–17] We limited our outcomes to those patients treated with antipsychotics given that benzodiazepines are used for many other indications (e.g., sedation) and

Table 1. Change Criteria Used to Identify Patients With Altered Mental Status[a]

Scale or Measure	Change Criteria	
	From	***To***
CAM-ICU	Negative	Positive
RASS	Score ≤1 (+1 [restless] to –5 [unarousable])	Score ≥2 (+2 [agitated] to +4 [combative])
Riker Sedation-Agitation Scale	Score ≤4 (4 [calm] to 1 [unarousable])	Score ≥5 (5 [agitated] to 7 [dangerous agitation])
Ramsay Sedation Scale	Score ≥2 (2 [cooperative, oriented, and tranquil] to 6 [no response to stimulus])	Score of 1 (anxious or restless, or both)
Orientation level[b]	Oriented	Disoriented
Consciousness level[b]	Alert	Irritable, lethargic, unresponsive, or other

[a]CAM-ICU = Confusion Assessment Method for the ICU, RASS = Richmond Agitation-Sedation Scale
[b]Orientation and consciousness levels were extracted from the electronic health record flow sheet.

antipsychotics are often used in more serious cases in which other management strategies have been unsuccessful. The administration of an antipsychotic medication had to occur within a time window between 2 hours before and 6 hours after the mental status change, allowing for documentation discrepancies or delays. We checked frequency and validity of AMS definitions with different time frames (i.e., 2, 4, and 6 hours within an antipsychotic medication administration) via empirical analysis and chart review to determine the window with optimized specificity and sensitivity. Additionally, the antipsychotic medication administered during this time window must not have been administered on the day prior to the mental status change to ensure it was given for treatment purposes (i.e., no scheduled administration).

AMS predictors. Medication-associated risk factors included the use of opioids, benzodiazepines, barbiturates, corticosteroids, or medications with anticholinergic effects (i.e., anticholinergics, antihistamines, or tricyclic antidepressants), total number of different ARIMs per day, cumulative medication days of ARIMs, and risk levels of ARIMs. We classified risk levels of ARIMs into low-, moderate-, and high-risk categories based on reported rates in their drug monographs. A medication with a rate of any type of AMS greater than 10% in either Clinical Pharmacology[18] or Micromedex[19] was classified into the high-risk category; a medication with AMS rate ranging from 1% to 10% either in Clinical Pharmacology or Micromedex was considered moderate risk; lastly, if the AMS rate was listed as less than 10% in either Clinical Pharmacology or Micromedex, we classified the medication as low risk. The categorization scheme and results were then reviewed by clinical pharmacists. The risk categorizations of ARIMs can be found in Appendix A.

More than 40 different nonmedication risk factors, including age, sex, underlying central nervous system pathology (e.g., brain tumor, central nervous system infection, history of delirium, hepatic encephalopathy, history of psychosis), other comorbidities (e.g., urinary tract infection, fracture, hearing loss, depression), and clinical signs and symptoms (e.g., increase in serum creatinine level, dehydration, elevated calcium level, hypoglycemia, hyponatremia), were identified from the published literature.[1,3-8] These risk factors were presented to a national technical expert panel including acute-care safety experts representing medicine, nursing, and pharmacy that supported the larger risk score development effort.[15-17] The panel ensured the list of risk factors was comprehensive and provided feedback on the ability to develop standardized measures of each from different EHRs and hospital systems across the country. Both the medication- and nonmedication-associated risk factors were defined to allow automated ascertainment from discrete fields in EHRs using patient demographics, vital signs, select billing codes (from the International Classification of Diseases, Ninth Revision, Clinical Modification code set), patient location, EHR assessment scales (e.g., pain, mobility, visual impairment), and inpatient laboratory values.

For each hospital day, we developed a set of risk factors derived from varying look-back periods to predict medication-associated AMS risk for the following day. Depending on their expected effect on the outcome, we allowed select risk factors to have a look-back period of up to 1 year, including information from inpatient and outpatient records (Appendix B).

We used missing data indicators or imputation for variables with missing EHR data.[20] For example, we used a missing indicator—meaning that no information on the variable was available for that patient—for partial pressure of carbon dioxide (P_{CO_2}) given its demonstrated predictive properties.

We imputed normal values for sodium and calcium serum concentrations if the values were missing with the assumption the variables would have normal values if tested. Similarly, if no vision or hearing assessment was recorded, we assumed patients were not visually or hearing impaired.

Statistical analyses. We first used univariate analyses to examine crude associations between risk factors and AMS events. Variables demonstrating unreliable measurement or no predictive value in the univariate analyses were excluded. Composite variables were created from risk factors infrequently observed that shared similar causal pathways. Next, we used cluster analyses to identify clusters until 75% of total variation was accounted for. Among each identified cluster, we selected the risk factor with the strongest predictive properties for inclusion in multivariate analyses. We then examined interactions between each remaining risk factor and hospital day (confined to the first 5 days in accordance with the study population) to observe whether the strength of association between the risk factor and AMS changed across hospital days. Because associations did not change appreciably (i.e., interactions were nonsignificant), we built a single prediction model by collapsing the 5 individual models into 1 model.

We used multivariate logistic regression to model the relationship between risk factors and AMS. We evaluated 3 different models: (1) a full model including all identified risk factors; (2) a parsimonious model developed via backward elimination technique that retained all variables with a p value <0.15; and (3) a reduced model with variables that were retained in at least 40 of 100 backward elimination processes that were repeated in 100 bootstrap samples. From these models, we used the Akaike information criterion to choose the best candidate model. The final model was validated using internal validation with bootstrapping. To correct for optimism resulting from using the same data for development and validation, we refitted the final model in 100 bootstrap samples of the original data set.[21] The resulting model parameters were then reapplied to the original data set, which yielded the optimism-adjusted C statistic as a model performance measure. Finally, to evaluate the clinical utility of the model, we estimated the proportion of observed AMS events among patients who had risk scores in the upper 50th and 90th percentiles. The risk score for each patient was calculated from the sum of beta values in the validated model.

The study was approved by the institutional review and privacy boards of the University of Florida.

Results

Among 83,787 patients admitted to the 2 study hospitals, we identified 66,875 who received at least 1 ARIM during the first 5 hospital days (80%). The mean ± S.D. length of stay among admitted patients with ARIM exposure was 3.19 ± 1.55 days (Table 2). Of the 194,156 hospital days with ARIM exposure (also referred to as modeling days herein), 262 were followed by a day with AMS (1.35 events per 1,000 patient-hospital days). The distribution of AMS events by hospital day was as follows: day 2 (i.e., the day after modeling day 1), 74; day 3, 59; day 4, 53; day 5, 48; and day 6, 28. The average age of ARIM-exposed patients was 54 years, with a slight majority of female patients (55%). Across the first 5 hospital days, 26.3% of patients were exposed to only 1 ARIM on a modeling day, 59.3% were exposed to 2–4 ARIMs on a modeling day, and 14.5% were exposed to >4 ARIMs on a modeling day.

All 3 models—full, parsimonious, and reduced—demonstrated similar discrimination, with unbiased C statistics of 0.86 for the full model and 0.85 for both backward-elimination models. The parsimonious model generated using the backward-elimination method was selected as the final model because it exhibited the lowest Akaike information criterion. The model demonstrated an acceptable optimism, suggesting minimal overfitting, with an unbiased C statistic of 0.85 and an original C statistic of 0.86 (95% confidence interval [CI], 0.84–0.89). The strongest predictors of AMS included a history of AMS (odds ratio [OR], 9.55; 95% CI, 5.64–16.17), alcohol withdrawal (OR, 3.34; 95% CI, 2.18–5.13), history of delirium or psychosis (OR, 3.25; 95% CI, 2.39–4.40), presence in the intensive care unit (ICU) (OR, 2.53; 95% CI, 1.89–3.39), and hypernatremia (OR, 2.40; 95% CI, 1.61–3.56). None of the medication-associated risk factors (i.e., medication classes, number of ARIMs, and ARIM risk level) were retained in the final model (Table 3).

Among patients scoring in the 90th percentile of risk, the model captured 159 medication-associated AMS events (60.7%). The upper 50th percentile of the risk score yielded 242 events (92.4%). Because of the low rate of AMS, among modeling days in the upper 90th and 50th percentile, we found 1 AMS event per 127 and 400 admission days, respectively (Figure 1).

Discussion

In this study, we developed and validated a medication-associated AMS predictive model. Although multiple risk models have been developed to detect AMS,[22–27] delirium particularly, previous risk models have not focused on medication-associated AMS. Additionally, prior models often rely on manual assessment, consider static admission characteristics alone, or are limited to specific demographics (e.g., elderly).[24,25] In contrast, our risk model considers these limitations in the following respects. First, the model was operationalized entirely by readily available EHR data, thus facilitating automation. Second, the model includes dynamic risk-factor information; that is, the different look-back periods and ability to update risk-factor information by modeling day accounted for changes in risk-factor values during a patient's hospital stay. A dynamic model is critical for accurate risk estimation, because some risk factors may have only short-term implications for risk or may

Table 2. Clinical Characteristics of the Study Population[a]

Characteristic	No. (%)[b]
Total study population	66,875 (...)[c]
Hospitalized days, mean ± S.D.	3.19 ± 1.55
Total patient days	194,156 (100)
Intensive care unit days	23,174 (11.94)
AMS events	262 (0.13)
Age, mean ± S.D. yr	53.66 ± 17.90
Male	88,253 (45.45)
Comorbid Conditions	
Alcohol withdrawal	4,471 (2.30)
Anemia	59,480 (30.64)
Delirium or psychosis history	15,500 (7.98)
Hypernatremia	3,644 (1.88)
Hypotension	45,261 (23.31)
Reduced mobility	74,984 (38.62)
Seizure	13,663 (7.04)
Thyroid dysfunction	24,623 (12.68)
Trauma	2,860 (1.47)
Visual impairment	4,943 (2.55)
Total Number of AMS Risk–Inducing Medication Exposures per Day	
1	50,940 (26.29)
2–4	115,106 (59.29)
≥5	28,110 (14.48)
AMS Risk–Inducing Medication Exposure by Classd	
Anticonvulsants	21,870 (11.26)
Benzodiazepines	39,895 (20.55)
Corticosteroids	32,186 (16.58)
Opioids	129,212 (66.55)
Selective serotonin reuptake inhibitors	26,305 (13.55)

[a]AMS = altered mental status.
[b]Percentages were calculated using the total patient days included in the study.
[c]Not applicable.
[d]A patient may have been exposed to more than one medication.

arise during the hospitalization. For example, while hypotension on admission would be a risk factor on day 1, it would not be a relevant risk factor on day 2 unless it persisted. Because the associations between risk factors and AMS were generally stable across the evaluated 5 hospital days (result not shown), we opted to build a single prediction model, but using modeling days as the unit of analysis retained our ability to update risk-factor information. Additionally, the pooling of hospital days allowed for the examination of more risk factors in the same model without risking overfitting.[28] Lastly, the model incorporated a broad range of outcome assessment scales not limited to delirium to accommodate a variety of inpatient settings.

Our model could accurately predict the risk of medication-associated AMS by using EHR predictors. A history of AMS during the hospital stay was the strongest predictor of AMS. Importantly, this variable is not the model's cornerstone, because removal of this variable from the model still yields a highly predictive model with an unbiased C statistic of 0.79. The predictors reflected in the model are consistent with the literature, with the exception of the missing-data indicator for $P\text{CO}_2$. Missing $P\text{CO}_2$ values had a protective effect, likely because patients with an arterial blood gas are often in critical condition. Thus, missing $P\text{CO}_2$ values in patients is a marker of their less critical condition.

The rate of medication-associated AMS in our study was 0.13% of cases per person-modeling day. This AMS rate and risk factors were specific to patients with ARIM exposures. The relatively low rate was a product of our AMS outcome definition, which necessitated pharmacologic treatment with antipsychotics. This outcome definition would not have captured cases managed nonpharmacologically (e.g., mild disorientation or hypoactive delirium) or cases characterized by agitation, in which a benzodiazepine was used as an acute treatment to reduce the potential of immediate harm to the patient or providers instead of an antipsychotic.[29] We limited our outcomes to those treated with antipsychotics given that benzodiazepines are used for many other indications (e.g., sedation) and antipsychotics are often used when other management strategies have been unsuccessful.

Medication-associated risk factors (i.e., total number of different ARIMs per day, cumulative ARIM medication-days, and risk levels of ARIMs) were not significant after adjustment for demographics, comorbidities, and other risk factors. The limited number of events hampered our ability to evaluate specific products and other medication-related factors, such as dosing (e.g., high-dose steroids), as AMS predictors. It should be noted here

Table 3. Adjusted Odds Ratios Between Risk Factors and Altered Mental Status[a]

Variable	Odds Ratio (95% Confidence Interval)
Risk Factors	
Age	1.03 (1.02–1.04)
Male	1.39 (1.08–1.80)
Presence in ICU	2.53 (1.89–3.39)
History of AMS	9.55 (5.64–16.17)
Trauma	2.24 (1.33–3.93)
Alcohol withdrawal	3.34 (2.18–5.13)
Anemia	1.32 (1.01–1.72)
History of delirium or psychosis	3.25 (2.39–4.40)
Hypotension	1.33 (1.03–1.73)
Hypernatremia	2.40 (1.61–3.56)
History of seizure	1.52 (1.01–2.29)
History of thyroid dysfunction	1.27 (0.93–1.75)
Missing P_{CO_2} value	0.37 (0.27–0.49)
Reduced mobility	1.88 (1.38–2.57)
Visual impairment	1.68 (0.97–2.91)
Hospital Day Indicators	
Day 1 (reference)	1.00
Day 2	0.83 (0.58–1.18)
Day 3	0.83 (0.57–1.22)
Day 4	0.86 (0.58–1.27)
Day 5	0.57 (0.36–0.91)

[a]AMS = altered mental status, ICU = intensive care unit, P_{CO_2} = partial pressure of carbon dioxide.

that all modeling days required at least 1 ARIM for inclusion, because our focus was to predict medication-associated AMS. Thus, the risk for AMS associated with exposure to ARIMs in general (relative to nonexposure) could not be examined here, and exclusion of ARIM-related risk factors from our risk model does not minimize their importance. However, it appears that the number, type, or AMS risk level of ARIMs did not play a major role in AMS risk after adjustment for other risk factors.

Some other opportunities for preventive pharmacist interventions emerged from our model results. Noteworthy, one of the strongest risk factors in the model was hypernatremia, which can be addressed with adequate fluid management. Likewise, hypotension appeared to be an independent risk factor even after adjustment for general frailty (e.g., anemia, trauma, presence in intensive care unit).

Our exclusion criteria limit the generalizability of our model to patients with baseline dementia or on mechanical ventilation as well as patients receiving continuous infusions of sedatives. We excluded these patients because of their inherently high risk of AMS. Given that dementia and delirium have substantial overlapping features and commonly coexist in the elderly, AMS in patients with dementia may not represent a reversible condition amenable to prevention.[30] Similarly, patients on mechanical ventilation or on continuous infusion of sedatives are at higher risk for AMS. Routine assessment for delirium, agitation, and sedation are recommended by various guidelines in addition to preventive measures in high-risk groups.[31,32] These recommendations have resulted in stringent protocols within the ICUs where patients in our study cohort received care. Including these patients in the model may have masked important factors for the general patient population. Therefore, we believe it is more appropriate to develop and validate a separate risk model for this cohort.

Although our risk model demonstrated excellent discriminatory ability, its immediate application to other hospitals would require validation and calibration. An individual hospital's protocols, formulary, EHR completeness and accurateness, and patient mix may affect the validity of the model. Similarly, the model's performance should be periodically optimized after implementation, because changes in the aforementioned risk factors and their measurement and documentation in the EHR can result in differences in risk model performance.

Conclusion

The model demonstrates that an automated evaluation of medication-associated AMS risk derived exclusively from EHR data is practical with excellent performance characteristics. The model represents important progress in automated clinical tools aimed at triaging finite resources to improve clinical outcomes.

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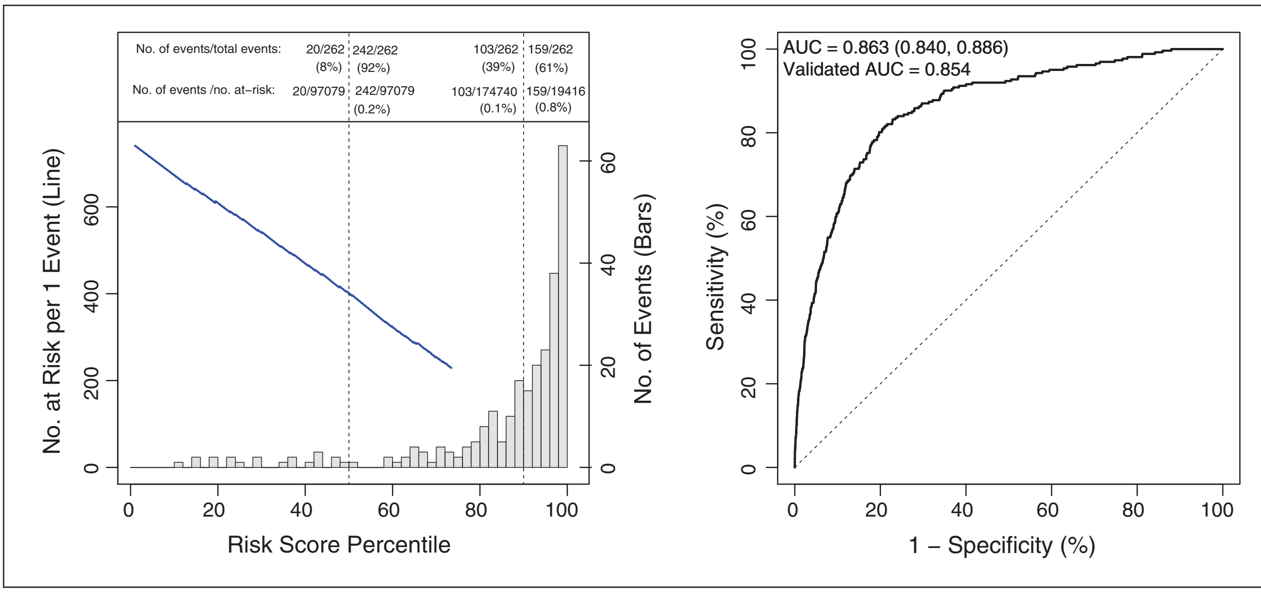
Figure 1. Left panel: Altered mental status rates and number at risk per 1 event by risk score percentile (at top of each panel) for the modeled risk period. The gray bars indicate the number of hospital days with events per risk score decile. The blue line represents the number of admissions flagged during a risk model day that would require intervention in order to prevent 1 AMS event on the subsequent day. Right panel: Area under the curve (AUC) values (with 95% confidence intervals) derived from area under the receiver operating characteristic curve analysis.



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Disclosures

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Additional information

This article reflects the views of the authors and should not be construed to represent the U.S. Food and Drug Administration's views or policies.

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Appendix A—Altered mental status risk–inducing medications

High risk (altered mental status [AMS] rate >10%)

- Aldesleukin
- Alprazolam
- Apomorphine
- Baclofen
- Bupropion
- Busulfan
- Citalopram
- Clonidine
- Corticotropin
- Fentanyl
- Hydrocodone
- Interferon alfa-2a
- Interferon alfa-2b
- Interferon alfacon-1
- Levetiracetam
- Lisdexamfetamine
- Mefloquine
- Oxymorphone
- Perampanel
- Pramipexole
- Ropinirole
- Rotigotine
- Tacrolimus
- Temazepam
- Tolcapone
- Topiramate
- Voriconazole
- Ziconotide

Moderate risk(AMS rate 1%–10 %)

- Acyclovir
- Amantadine
- Armodafinil
- Atomoxetine
- Atropine
- Dexmethylphenidate
- Dronabinol
- Entacapone
- Eszopiclone
- Ezogabine
- Fluoxetine
- Fluvoxamine
- Fosphenytoin
- Hydromorphone
- Memantine
- Methylphenidate
- Mirtazapine
- Morphine
- Nabilone
- Oxcarbazepine
- Oxybutynin
- Oxycodone
- Paroxetine
- Prednisolone
- Prednisone
- Pregabalin
- Promethazine
- Rasagiline
- Selegiline
- Sertraline
- Tapentadol
- Tizanidine
- Tramadol
- Verapamil

Low risk (AMS rate <1%)

- Buprenorphine
- Butabarbital
- Ciprofloxacin
- Dextroamphetamine
- Doxepin
- Efavirenz
- Estazolam
- Hypericum
- Levofloxacin
- Lorazepam
- Midazolam
- Modafinil
- Moxifloxacin
- Nifedipine
- Ofloxacin
- Pentobarbital
- Phenobarbital
- Secobarbital
- Zolpidem

Rate not reported (AMS described in product labeling, but percent not provided)

- Amiodarone
- Amitriptyline
- Amobarbital
- Atovaquone
- Benzphetamine
- Benztropine
- Betamethasone
- Bromocriptine
- Brompheniramine
- Cabergoline
- Carbamazepine
- Chloramphenicol
- Chlordiazepoxide
- Chloroquine
- Chlorphenamine
- Chlorpromazine
- Cimetidine
- Clomiphene
- Clonazepam
- Clorazepate
- Cortisone
- Cycloserine
- Desipramine
- Dexamethasone
- Dextromethorphan
- Diazepam
- Diethylpropion
- Digoxin
- Dimenhydrinate
- Diphenhydramine
- Disulfiram
- Doxylamine
- Droxidopa
- Ephedra
- Ephedrine
- Ergoloid
- Escitalopram
- Famotidine
- Fludrocortisone
- Flurazepam
- Gatifloxacin
- Gemifloxacin
- Guanidine
- Hydrocortisone
- Hydroxyzine
- Imipramine
- Interferon alfa-N1
- Interferon alfa-N3
- Interferon beta-1a
- Interferon beta-1b
- Isocarboxazid
- Isotretinoin
- Ketamine
- Levodopa
- Lidocaine
- Lithium
- Lomefloxacin
- Meclizine
- Meperidine
- Mephobarbital
- Metamphetamine
- Methadone
- Methohexital
- Methsuximide
- Methyldopa
- Methylprednisolone
- Micafungin
- Mycophenolate
- Naloxone

- Norfloxacin
- Nortriptyline
- Oseltamivir
- Oxazepam
- Pemoline
- Peramivir
- Phendimetrazine
- Phenelzine
- Phenytoin
- Pilocarpine
- Primidone
- Procainamide
- Propoxyphene
- Quinidine
- Ranitidine
- S-Adenosyl
- Scopolamine
- Sulindac
- Suvorexant
- Theophylline
- Thiopental
- Thioridazine
- Tranylcypromine
- Triamcinolone
- Trihexyphenidyl
- Trimipramine
- Trovafloxacin
- Warfarin

Appendix B—Operational definitions of variables included in multivariate analyses[a]

Variable	Definition	Look-Back Period (max)
Age	Age calculated as of day of admission	None (if using provided age); 365 days (if using date of birth)
Sex	Sex recorded in admission record	None
Visual impairment[b]	ICD 9-CM: 369 (blindness and low vision), 369.9 (unspecified visual loss), 368.16 (psychophysical visual disturbances), and 438.7 (disturbances of vision)	1 yr
Alcohol withdrawal	Multivitamin/thiamine/folic acid infusion, *or* acamprosate on modeling day, *or* CIWA total score ≥1 at or up to 3 days prior to modeling day	None (infusion); 3 days (CIWA)
Anemia	Hemoglobin level <10	1 day
Hypotension	The lowest value at modeling day is <80 mmHg for SBP or <65mmHg for MAP at or up to 1 day before modeling day (use MAP = [2 DBP + SBP]/3); if missing, impute as nonpresence of hypotension	1 day
History of delirium or psychosis[b]	ICD 9-CM: 780.09 (acute delirium), 291.0 (alcoholic withdrawal delirium), 293.89 (chronic delirium), 292.81 (drug-induced delirium), 293.1 (subacute delirium), 293.0 (delirium due to conditions classified elsewhere), 298.9 (psychosis), 298.1 (acute hysterical psychosis), 295.0–295.9 (schizophrenia [all])	1 yr
Seizure disorders[b]	ICD 9-CM 345.xx (all)	1 yr
Hypothyroidism[b]	ICD 9-CM: 244 (acquired hypothyroidism), 244.0 (postsurgical hypothyroidism), 244.1 (other postablative hypothyroidism), 244.2 (iodine hypothyroidism), 244.3 (other iatrogenic hypothyroidism), 244.8 (other specified acquired hypothyroidism), 244.9 (unspecified acquired hypothyroidism), and 243 (congenital hypothyroidism)	1 yr
$P\text{CO}_2$ not measured	$P\text{CO}_2$ not measured on modeling day	None
Hypernatremia	Sodium >145 mg/dL	None
Presence in ICU	Patient spent any amount of time in an ICU on the modeling day	None
Reduced mobility	Flag if the following occurs: on the modeling day, at least one record of "bed in chair position," "bedrest," "chair," "dangle," "dangled," "in bed," "total care sport," "turn," or "up in wheelchair" must be present, and *none* of the following may be present: "ambulate(d) in hall," "ambulate(d) in room," "ambulated to bathroom," "assistance with relearning ADL activities," "bathroom privileges," "commode," "does not require skilled therapy services," "other (comment)," "stand(stood) at bedside," "stretcher," "therapy consulted via md order," "up ad lib," "up in chair," "up in stretcher chair"	3 days
Trauma	Patient entered the operating room within 8 hr of being admitted to the emergency department, and patient was admitted to the hospital on or before exiting the operating room	1 day

[a] CIWA = Clinical Institute Withdrawal Assessment for Alcohol, SBP = systolic blood pressure, MAP = mean arterial pressure, DBP = diastolic blood pressure, ICD 9-CM = International Classification of Diseases Ninth Revision, Clinical Modification, $P\text{CO}_2$ = partial pressure of carbon dioxide, ICU = intensive care unit.
[b] Present on admission or up to 365 days prior to admission.